Theoretical Foundations for Uncertainty-Aware Simplex-Based Open-Set Recognition: A General Framework Beyond the Simplex Regime

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Abstract

In medical imaging and other safety-critical settings, classifiers must not only recognize known pathologies but also confidently reject unseen conditions, a capability formalized as open-set recognition (OSR). Recent simplex-based methods like the Deep Simplex Classifier and its uncertainty-aware variant achieve strong empirical results by aligning class prototypes with a regular simplex geometry and using a distance-ratio score for rejection. Yet despite their success in practice, these methods have lacked a rigorous theoretical foundation, and existing analyses have been restricted to the regime $d \geq C - 1$ where a regular simplex can be realized.

This paper develops a unified theoretical framework for simplex-ratio OSR that works in *all* embedding dimensions, including the practically important low-dimensional regime $d < C - 1$. For arbitrary distinct prototypes, we prove that the uncertainty score U is globally Lipschitz, that acceptance regions are compact, and that U-sublevel sets are sandwiched between explicit ball unions. For the weaker class of *balanced equal-norm codes*—which exist in every dimension $d \geq 2$—we show that the auxiliary squared score U_2 has sublevel sets that are exact compact unions of Euclidean balls and establish a sharp dichotomy: perfect one-distance simplex symmetry holds if and only if $d \geq C - 1$, and below this threshold the core rejection geometry survives with explicit degradation governed by a simplex-defect parameter λ_j. We derive false acceptance rate bounds that scale exponentially with embedding dimension under natural isotropy assumptions, analyze the interplay between the three UCDSC loss components, and provide practical algorithms for data-driven threshold selection. We additionally complement the theory with focused experiments on reproduced UCDSC baselines for BloodMNIST and DermaMNIST. These experiments show that recall-controlled thresholding induces a strong FAR–KRR trade-off, that reducing the embedding dimension from the default 512 to 32 on BloodMNIST degrades AUROC and OSCR while substantially worsening the best-DTACC operating point, and that empirical $\hat{\delta}_{\text{cap}}$ estimates on the reproduced runs remain small and similar across datasets. Thus the experiments support the practical relevance of the threshold and dimension analyses, while leaving the anisotropy failure mode suggestive but not yet cleanly isolated empirically.

Keywords: Open-set recognition, simplex ETF, neural collapse, uncertainty quantification, theoretical guarantees, medical imaging, balanced equal-norm codes, low-dimensional embeddings.

1 Introduction

Open-set recognition (OSR) formalizes the practical scenario where a classifier must simultaneously identify samples from known training classes and reject inputs from previously unseen classes at test time [1, 2]. Unlike closed-set classification, which assumes the label space is shared between training and testing, OSR acknowledges the open-world nature of real deployments and requires explicit mechanisms for novelty detection [3].

This capability is especially critical in medical imaging, where rare diseases, imaging artifacts, and previously unidentified pathologies can arise at any time [4]. A model that confidently assigns a known-class label to a truly novel input can lead to misdiagnosis with severe clinical consequences. Consequently, there is growing interest in OSR methods that are both accurate and theoretically well-founded for safety-critical applications.

Neural collapse and simplex classifiers. Recent theoretical insights reveal that deep neural classifiers, during the terminal phase of training, exhibit a striking geometric convergence known as *neural collapse* (NC) [5]: last-layer features collapse to their class means, and these means align with the vertices of a simplex equiangular tight frame (ETF). This observation has motivated a family of classifiers that fix the class prototypes at simplex ETF vertices from the outset, simultaneously maximizing both Euclidean and angular inter-class margins [6].

The Deep Simplex Classifier (DSC) [6] trains features to cluster around fixed simplex vertices using an intra-class loss, with a triplet-based outlier-exposure term for

background rejection. The Uncertainty-aware Deep Simplex Classifier (UCDSC) [7] extends DSC by adding an uncertainty-aware regularization term L_u, inspired by the distance-ratio scoring of Cao et al. [23], that penalizes the open space between class prototypes. UCDSC achieves state-of-the-art performance on multiple MedMNIST benchmarks [8] and the Augmented Skin Conditions dataset [9].

The theory gap. Despite these strong empirical results, neither DSC nor UCDSC provides formal theoretical analysis. Fundamental questions remain unanswered: *Why does the simplex-ratio score U work for open-set detection? Under what conditions is the acceptance region compact? How does the false acceptance rate (FAR) scale with embedding dimension? Why does the L_u term help even without background data? How should the threshold θ be chosen to guarantee a target known-class recall?* Moreover, existing simplex-based methods assume $d \geq C-1$ in order to realize a regular simplex, yet practical embedding dimensions can be small relative to the number of classes. *Does the rejection geometry collapse when $d < C-1$, or do the core guarantees survive?*

Our contributions. We develop a unified theoretical framework for simplex-ratio OSR that works for *all* prototype configurations, not only the regular simplex, and that covers *all* embedding dimensions including $d < C-1$. Our main contributions are:

1. **General Lipschitz analysis and ball control (Section 4).** For *arbitrary distinct* prototypes (no simplex assumption), we establish global Lipschitz continuity of U with constant $4C/((C-1)\Delta)$ where Δ is the minimum inter-prototype distance, and prove that U-acceptance regions are compact finite unions of prototype-centered balls with explicit inner and outer radii (Theorem 2). For the regular-simplex special case, the Lipschitz constant sharpens to $4/D$. Under assumption A5, we derive class-conditional quantile bounds enabling population recall control (Theorem 12(i)).

2. **Exact U_2 geometry for balanced equal-norm codes (Section 5).** We introduce *balanced equal-norm codes* ($\|s_j\| = R$, $\sum_j s_j = 0$), which exist in every dimension $d \geq 2$ and include regular simplices as a special case. For these codes, we characterize U_2-sublevel sets as exact unions of at most C closed Euclidean balls with explicit centers and radii, and show that U-acceptance regions lie sandwiched between corresponding ball unions (Theorem 8).

3. **Sharp dichotomy and defect parameter (Section 5.1).** We introduce the simplex-defect parameter $\lambda_j = \sqrt{B_j}/A_j \geq 1$, where A_j and B_j are the mean and mean-squared rival distances from class j. We prove a sharp dichotomy: $\lambda_j = 1$ for all j if and only if the prototypes form a regular simplex, which requires $d \geq C-1$ (Theorem 10). Below this threshold, we establish a local equivalence $U \asymp \sqrt{U_2}$ with explicit constants depending on λ_j (Theorem 11).

4. **Concrete low-dimensional construction (Section 5.2).** We provide a universal construction—the regular C-gon embedded in $\mathbb{R}^d$ for any $d \geq 2$—with fully explicit formulas for all geometric statistics, proving that the framework is non-vacuous in every low-dimensional regime.

5. **Conditional FAR bounds (Section 6).** Under normalized unknown directions, we prove a FAR upper bound of the form $C\exp(-c(\theta)d) + C\,\delta_{\text{cap}}$ and, under exact isotropy, a conservative spherical-cap lower bound. Under the radial-isotropic unknown-feature model, we prove an exponential-in-d FAR upper bound (Theorem 12(iii)–(iv)).

6. **UCDSC loss analysis (Section 7).** We show that the three UCDSC loss components play complementary but different roles: Proposition 13 controls the average training score, Theorem 16 gives a conditional empirical separation guarantee for training knowns versus auxiliary background samples, and Proposition 15 together with Corollary 20 provide a local negative-gradient interpretation for L_u.

7. **Practical corollaries (Section 8).** We derive closed-form guidance for threshold selection, expand-factor effects, and a sufficient embedding dimension for a target FAR. We also provide a concrete, validation-based proxy for estimating the rescaled score-tail parameter σ from data (Algorithm 1), making the population threshold rule operational up to the usual finite-sample caveats.

8. **Focused empirical support on reproduced UCDSC baselines (Section 9).** We reproduce UCDSC baselines on BloodMNIST and DermaMNIST, quantify the FAR–KRR tradeoff induced by recall-controlled thresholding, and run a multi-split low-dimension comparison on BloodMNIST. These experiments support the practical relevance of the threshold and dimension analyses, while showing that the anisotropy warning is not yet cleanly isolated empirically.

Main Results

- **Global regularity (any distinct prototypes).** The uncertainty score $U(z) = d_{\min}(z)/\mu(z)$ is globally Lipschitz: $|U(z) - U(z')| \leq \frac{4C}{(C-1)\Delta}\|z - z'\|$; acceptance regions $\{U \leq \theta\}$ are compact finite unions of prototype-centered balls with explicit radii. *Theorem 2*
- **Exact U_2 geometry (balanced equal-norm codes).** For $\|s_j\| = R$, $\sum_j s_j = 0$: $\{U_2 \leq \rho\} = \bigcup_{j=1}^{C} \mathbb{B}_j(\rho)$, a union of at most C Euclidean balls with explicit centers/radii; for U, matching inner/outer ball bounds hold. Works in *every* dimension $d \geq 2$. *Theorem 8*
- **Sharp dichotomy.** $\lambda_j = 1$ for all $j \iff$ prototypes form a regular simplex $\iff d \geq C - 1$. Below this threshold, the defect $\max_j \lambda_j > 1$ quantifies graceful degradation, not collapse. *Theorem 10*
- **Conditional FAR bounds.** Under the normalized-route model in A6/A6′, $\mathrm{FAR}(\theta) \leq C\exp(-\frac{(d-1)t(\theta^2)^2}{2}) + C\,\delta_{\mathrm{cap}}$; under exact isotropy this yields exponential decay of the upper bound in d, and under the radial-isotropic model the FAR upper bound is exponential in d. *Theorem 12*
- **Training/background score separation.** Proposition 13 controls the average training U-score; if $L_o = 0$, Theorem 16 gives deterministic score bounds for the realized training known and auxiliary background features, and a separating threshold exists whenever the known features are uniformly close enough to their prototypes. Proposition 15 and Corollary 20 provide a local negative-gradient mechanism for L_u.

Paper structure. Section 2 reviews related work. Section 3 states the formal setup, introducing the hierarchy of prototype assumptions (A1/A1′/A1″) and connecting the two score variants (U and U_2). Section 4 establishes Lipschitz analysis and ball control for general prototypes. Section 5 develops exact U_2 geometry under balanced equal-norm codes, the sharp dichotomy theorem, local comparison bounds, and the C-gon construction. Sections 6–8 develop FAR bounds, loss analysis, and practical corollaries. Section 9 reports focused reproduced experiments. Section 10 concludes. Full proofs appear in the Appendix.

2 Related Work

Open-set recognition. The foundations of OSR were laid by Scheirer et al. [1], who formalized open-space risk and later proposed compact abating probability (CAP) models [10]. Subsequent work explored diverse rejection mechanisms: OpenMax [11] adapts extreme-value calibration to softmax outputs, DOC [12] uses per-class sigmoid thresholds, and reconstruction-based approaches like CROSR [13] and C2AE [14] leverage reconstruction error as a novelty signal. Distance-based methods represent another strand, including NNO [15], OSNN [16], EVM [17], and class anchor clustering [31].

Prototype and simplex methods. More recent work has turned toward prototype-based approaches for their interpretability and geometric structure. GCPL [18] learns class prototypes with an explicit loss, while RPL [19] introduces reciprocal points to explicitly model extra-class space. This idea was refined by ARPL [20], which adds adversarial margin constraints. In parallel, methods like DIAS [21] and OMCL [22] use feature generation and angular margins to control open space. More directly relevant to this work, DSC [6] fixes prototypes at simplex ETF vertices and employs intra-class clustering with triplet-based outlier exposure [29]. UCDSC [7] extends this by adding an uncertainty-aware regularization that penalizes high inter-prototype distances.

Neural collapse and simplex geometry. The simplex structure underlying DSC and UCDSC connects to the broader phenomenon of neural collapse [5], which describes how features and classifiers converge to simplex ETF geometry during training. Subsequent theoretical work [24–27] has characterized neural collapse under various conditions. Our analysis differs in scope: rather than explaining why training produces simplex geometry, we provide guarantees for the *downstream open-set recognition* task that simplex-aligned embeddings enable.

Medical imaging OSR. OSR applications in medical imaging have grown in importance, with work exploring MedMNIST benchmarks [8], evidential uncertainty quantification [4], and distillation-based improvements [28]. However, no prior work establishes theoretical guarantees specific to simplex-based OSR in medical or clinical settings.

3 Preliminaries

We state the formal assumptions underpinning our framework and fix notation used throughout.

Notation and score variants. We write $[C] := \{1, \ldots, C\}$, $[x]_+ := \max(x, 0)$, and use $\mathbb{P}$ and $\mathbb{E}$ for probability and expectation. Both UCDSC training and the reject rule operate on the unsquared distance-ratio score $U(z) = d_{\min}(z)/\mu(z)$. We show that U is globally Lipschitz for *any* distinct prototypes (Section 4). For geometric tractability, we additionally introduce a squared variant $U_2(z)$ (Assumption 5) whose sublevel sets admit closed-form descriptions under the balanced equal-norm condition A1′ (Section 5). Proposition 6 establishes the global lower bound $\sqrt{U_2(z)} \leq U(z)$. Under the balanced equal-norm condition A1′, Theorem 8(iv) yields the global sandwich $\{U_2 \leq \theta^2/2\} \subseteq \{U \leq \theta\} \subseteq \{U_2 \leq \theta^2\}$; under the simplex (A1″), Lemma 7 improves the inner constant to $3\theta^2/4$. Near each prototype, Theorem 11 further sharpens the comparison with explicit dependence on the simplex-defect parameter λ_j.

Assumption 1 (A0: Basic objects)**.** Let $C \geq 2$ be the number of known classes and $d \geq 2$ the embedding di-

mension. Let $f : \mathcal{X} \to \mathbb{R}^d$ be the learned embedding. The known distribution is P_K on $(X, Y) \in \mathcal{X} \times [C]$, with $Z := f(X)$. The unknown distribution is P_U on $X \in \mathcal{X}$, with $Z_U := f(X)$. The auxiliary background distribution is P_B on $X_B \in \mathcal{X}$, with $Z_B := f(X_B)$.

We introduce a hierarchy of prototype assumptions of increasing strength. The weakest (A1) requires only distinct prototypes and suffices for Lipschitz regularity and compact acceptance regions. The intermediate (A1′) adds balanced equal-norm structure and enables exact U_2 ball geometry. The strongest (A1″) recovers the regular simplex as a special case.

Assumption 2 (A1: Distinct prototypes)**.** Prototypes $\{s_1, \dots, s_C\} \subset \mathbb{R}^d$ are distinct. Define the geometric statistics:

$$A_j := \frac{1}{C-1} \sum_{k \neq j} |s_j - s_k|, \quad B_j := \frac{1}{C-1} \sum_{k \neq j} |s_j - s_k|^2, \tag{1}$$

$$\Delta_j := \min_{k \neq j} |s_j - s_k|, \qquad \lambda_j := \frac{\sqrt{B_j}}{A_j} \geq 1. \tag{2}$$

Set $A_{\min} := \min_j A_j$, $A_{\max} := \max_j A_j$, $\Delta := \min_j \Delta_j$. We call λ_j the *simplex-defect parameter* of class j: it equals 1 if and only if all rival distances from s_j are equal.

Assumption 3 (A1′: Balanced equal-norm code)**.** The prototypes satisfy A1 and additionally:

$$\|s_j\| = R \quad \forall j, \qquad \sum_{j=1}^{C} s_j = 0, \tag{3}$$

for some $R > 0$. Such codes exist in every dimension $d \geq 2$ for every $C \geq 3$: a regular C-gon in a 2-plane embedded in $\mathbb{R}^d$ suffices (see Section 5.2).

Assumption 4 (A1″: Simplex-ETF prototypes)**.** The prototypes satisfy A1′ and additionally form a regular simplex:

$$\langle s_i, s_j \rangle = -\frac{R^2}{C-1} \quad (i \neq j). \tag{4}$$

The common edge length is $D := \|s_i - s_j\| = R\sqrt{2C/(C-1)}$. This requires $d \geq C - 1$. Under A1″, $A_j = D$, $B_j = D^2$, $\Delta_j = D$, and $\lambda_j = 1$ for all j.

Assumption 5 (A2: Scores)**.** For $z \in \mathbb{R}^d$, let $d_j(z) := \|z - s_j\|$, $d_{\min}(z) := \min_j d_j(z)$, $j^*(z) := \arg\min_j d_j(z)$ (with ties broken by choosing the smallest index), and

$$\mu(z) := \frac{1}{C-1} \sum_{k \neq j^*(z)} d_k(z), \qquad U(z) := \frac{d_{\min}(z)}{\mu(z)}, \tag{5a}$$

$$\nu(z) := \frac{1}{C-1} \left(\sum_j \|z - s_j\|^2 - \min_j \|z - s_j\|^2 \right), \tag{5b}$$

$$U_2(z) := \frac{\min_j \|z - s_j\|^2}{\nu(z)}, \qquad \alpha(z) := \max_j \langle z, s_j \rangle. \tag{5c}$$

Note that U_2 is a distinct score, not the square of U.

Tie-breaking. When the minimizer is not unique, we define $j^*(z) := \min \arg\min_{j \in [C]} d_j(z)$ (smallest index among minimizers). *Remark:* ties occur only on a finite union of bisector hyperplanes and hence on a Lebesgue-null set; this convention does not affect any probabilistic statements under absolutely continuous distributions (which includes all models considered in this paper).

Assumption 6 (A3: Reject rule)**.** Accept as known iff $U(z) \leq \theta$ for threshold $\theta \in [0, 1]$. The closed-set label is $\hat{y}(z) := j^*(z)$.

Assumption 7 (A4: UCDSC loss)**.** Given training data $\{(x_i, y_i)\}_{i=1}^n$ with $y_i \in [C]$ and background samples $\{x_k^{\text{bg}}\}_{k=1}^K$, define $f_i := f(x_i)$ and $f_k^{\text{bg}} := f(x_k^{\text{bg}})$. Let $m > 0$ be a margin parameter and $\lambda_o, \lambda_u \geq 0$ be loss weights. The total loss is $L_{\text{total}} = L_{\text{intra}} + \lambda_o L_o + \lambda_u L_u$ where:

$$L_{\text{intra}} := \frac{1}{n} \sum_{i=1}^{n} \|f_i - s_{y_i}\|^2, \tag{6}$$

$$L_o := \sum_{i=1}^{n} \sum_{k=1}^{K} \max\big(0, m + \|f_i - s_{y_i}\|^2 - \|f_k^{\text{bg}} - s_{y_i}\|^2\big), \tag{7}$$

$$L_u := \frac{1}{n} \sum_{i=1}^{n} U(f_i). \tag{8}$$

Norm convention. Note that U in (5a) (and hence L_u) uses *unsquared* Euclidean norms $\|z - s_j\|$, matching the UCDSC definition [7], whereas L_{intra} (6) and L_o (7) use squared norms $\|z - s_j\|^2$.

Assumption 8 (A5: One-sided score-tail control)**.** For each class y, the uncertainty score $U(Z) \mid Y = y$ satisfies a one-sided sub-Gaussian upper-tail inequality about its mean with scale τ_y: for all $t \geq 0$,

$$\mathbb{P}(U(Z) - \mathbb{E}[U(Z) \mid Y = y] \geq t \mid Y = y) \leq \exp\left(-\frac{t^2}{2\tau_y^2}\right), \tag{9}$$

with $\tau := \max_y \tau_y$. We define $\sigma_y := \tau_y / L_U$ where L_U is the Lipschitz constant of U: under A1 alone, $L_U = 4C/((C-1)\Delta)$; under A1″ (simplex), $L_U = 4/D$. Set $\sigma := \max_y \sigma_y$, so that $\tau = L_U \sigma$. Since $U(Z) \in [0, 1]$, Hoeffding's lemma yields (9) with the universal choice $\tau_y = 1/2$. In the results below, A5 is used only through whatever sharper class-specific values of τ_y are separately assumed or validated from data.

Remark 1 (Scope of A5). A5 is a condition on the *scalar* score $U(Z) \mid Y = y$. The paper does not claim that simplex geometry or UCDSC training alone implies A5. If one has an independent feature-space concentration theorem for $Z \mid Y = y$ that transfers to every L-Lipschitz observable g, then applying it to $g = U$ (with $L = L_U$) yields (9) with $\tau_y = L_U \sigma_y$. We do not rely on any specific such theorem here.

Assumption 9 (A6: Unknown features)**.** Two analysis routes, each specifying a pre-processing map Φ: **(U-Norm):** $\Phi(z) = Rz/\|z\|$ for $z \neq 0$, and $\Phi(0) := 0$ by convention (this is immaterial for the FAR bounds below, which are stated for $\theta < 1$, because then $U(0) = 1$ and the origin is rejected); assume that for all unit vectors $v \in S^{d-1}$ and all $t \in [0,1]$, $|\mathbb{P}(\langle \Phi(Z_U)/R, v\rangle \geq t \mid Z_U \neq 0) - p_{\mathrm{cap}}(t;d)| \leq \delta_{\mathrm{cap}}$, where $\delta_{\mathrm{cap}} \geq 0$ is a fixed constant. **(U-RadIso):** $\Phi = \mathrm{id}$ (no normalization); write $Z_U = R_U V_U$ with $R_U := \|Z_U\|$ and $V_U \mid R_U$ uniform on S^{d-1}.

Assumption 10 (A6′: Cap tail)**.** $p_{\mathrm{cap}}(t;d) := \mathbb{P}(\langle W, v\rangle \geq t) \leq \exp(-(d-1)t^2/2)$ for any fixed $v \in S^{d-1}$, $t \in [0,1]$, and $W \sim \mathrm{Unif}(S^{d-1})$. The quantity is independent of v by rotational symmetry.

We define the operating metrics (where Φ is specified by the chosen model in A6):

$$\mathrm{FAR}(\theta) := \mathbb{P}(U(\Phi(Z_U)) \leq \theta), \quad \mathrm{KRR}(\theta) := \mathbb{P}(U(Z) > \theta). \tag{10}$$

4 Lipschitz Analysis, Ball Control, and Concentration

We begin by establishing results for *arbitrary distinct* prototypes (A1 only), requiring no simplex or balanced-code structure. We then specialize to the simplex (A1″) for sharper constants.

Theorem 2 (Generic ball control and Lipschitz stability)**.** *Assume only A1 (distinct prototypes) and A2. Then for every $\theta \in (0,1)$:*

$$\bigcup_{j=1}^{C} B\big(s_j,\ r_j^-\big) \subseteq \{U \leq \theta\} \subseteq \bigcup_{j=1}^{C} \overline{B}\big(s_j,\ r_j^+\big)\,,$$

where

$$r_j^- := \min\left\{\frac{\Delta_j}{2},\ \frac{\theta A_j}{1+\theta}\right\}, \qquad r_j^+ := \frac{\theta A_j}{1-\theta}.$$

In particular $\{U \leq \theta\}$ is compact. Moreover, U is globally Lipschitz:

$$|U(z) - U(w)| \leq \frac{4C}{(C-1)\Delta}\|z-w\|, \qquad \Delta := \min_{i\neq j} |s_i - s_j|. \tag{11}$$

Proof. Let $z \in \mathbb{R}^d$, $j = j^*(z)$, $r = d_j(z) = d_{\min}(z)$.

Outer bound. For every $k \neq j$, triangle inequality gives $d_k(z) \leq r + |s_j - s_k|$. Averaging over $k \neq j$: $\mu(z) \leq r + A_j$. Hence $U(z) = r/\mu(z) \geq r/(r + A_j)$. So if $U(z) \leq \theta$, then $r/(r + A_j) \leq \theta$, giving $r \leq \theta A_j/(1-\theta) = r_j^+$.

Inner bound. Fix j and suppose $|z - s_j| < \Delta_j/2$. Then for every $k \neq j$, $d_k(z) \geq |s_k - s_j| - |z - s_j| > \Delta_j/2 > |z - s_j|$, so j is the unique nearest prototype. Therefore $\mu(z) \geq A_j - |z - s_j|$, hence $U(z) \leq |z - s_j|/(A_j - |z - s_j|)$. If $|z - s_j| \leq \theta A_j/(1+\theta)$, the right side is at most θ.

Lipschitz bound. $d_{\min}$ is 1-Lipschitz. $\mu(z) = \frac{1}{C-1}(\sum_m d_m(z) - d_{\min}(z))$, so μ is $\frac{C+1}{C-1}$-Lipschitz. For any $k \neq j^*(z)$: $d_k(z) + d_{\min}(z) \geq |s_k - s_{j^*}| \geq \Delta$, so $d_k(z) \geq \Delta/2$, hence $\mu(z) \geq \Delta/2$. The quotient rule gives (11). □

Under the simplex assumption A1″, this sharpens considerably:

Lemma 3 (Lipschitz continuity under A1″)**.** *Under A1″–A2: (i) $d_k(z) \geq D/2$ for all $k \neq j^*(z)$; in particular, $\mu(z) \geq D/2$; (ii) $|U(z) - U(z')| \leq (4/D)\|z - z'\|$ for all z, z'.*

Proof sketch. Under A1″, all pairwise distances equal D. The identity $\mu = \max_j \mu_j$ (where $\mu_j := \frac{1}{C-1}\sum_{k\neq j} d_k$) shows μ is exactly 1-Lipschitz, improving the $\frac{C+1}{C-1}$ factor. The quotient rule with denominator $\geq D/2$ then gives $L_U = 4/D$. Full proof in Appendix A. □

Remark 4 (Tightness). For $C = 2$, the bound $4/D$ is tight: $U(z) = r/(D-r)$ along the line joining the two prototypes has one-sided derivative $D/(D-r)^2 = 4/D$ at the midpoint. The general-prototype constant $4C/((C-1)\Delta)$ is not sharp in the simplex case: when all pairwise distances equal D, it reduces only to $4C/((C-1)D)$. Lemma 3 improves this to $4/D$ by exploiting the simplex identity $\mu = \max_j \mu_j$.

Quantile convention. For a real-valued random variable W, write

$$q_{1-\varepsilon}(W) := \inf\{t \in \mathbb{R} : \mathbb{P}(W \leq t) \geq 1 - \varepsilon\}.$$

Conditional quantiles are defined analogously.

Lemma 5 (Known-class quantile control)**.** *Under A1–A2 and A5: for each class y and $\varepsilon \in (0,1)$,*

$$q_{1-\varepsilon}(U(Z) \mid Y = y) \leq \mathbb{E}[U(Z) \mid Y = y] + \tau_y\sqrt{2\log(1/\varepsilon)}. \tag{12}$$

In particular, using $\tau_y = L_U\sigma_y$ and $\tau = L_U\sigma$:

$$\begin{aligned} q_{1-\varepsilon}(U(Z) \mid Y = y) &\leq \mathbb{E}[U(Z) \mid Y = y] + L_U\sigma_y\sqrt{2\log(1/\varepsilon)} \\ &\leq \mathbb{E}[U(Z) \mid Y = y] + L_U\sigma\sqrt{2\log(1/\varepsilon)}. \end{aligned}$$

Proof. Invert the A5 tail bound (9) at level ε. □

5 Exact U_2 Geometry and Bounds for U

We now show that the auxiliary score U_2 has an exact union-of-balls geometry under the balanced equal-norm condition (A1′), and that the operational acceptance region $\{z : U(z) \leq \theta\}$ lies between corresponding unions of Euclidean balls. We first establish a global comparison between U and U_2 that holds for *all* prototype configurations.

Proposition 6 (Global lower comparison)**.** *Under A1–A2 (any distinct prototypes), for every z:*

$$\sqrt{U_2(z)} \le U(z). \tag{13}$$

Hence $\{U \le \theta\} \subseteq \{U_2 \le \theta^2\}$ for every $\theta \ge 0$.

Proof. Let $j = j^*(z)$, $m = d_j(z)$, and $a_k = d_k(z)$ for $k \ne j$. Write

$$\mathrm{AM} := \frac{1}{C-1}\sum_{k\ne j} a_k = \mu(z), \qquad \mathrm{RMS} := \sqrt{\frac{1}{C-1}\sum_{k\ne j} a_k^2} = \sqrt{\nu(z)}.$$

Then $U(z) = m/\mathrm{AM}$ and $\sqrt{U_2(z)} = m/\mathrm{RMS}$, while $\mathrm{AM} \le \mathrm{RMS}$ by Cauchy–Schwarz. □

Under the simplex (A1″), we additionally have a tight *upper* comparison:

Lemma 7 (U–U_2 equivalence under A1″)**.** *Under A1″–A2, for all $z \in \mathbb{R}^d$:*

$$\sqrt{U_2(z)} \le U(z) \le \frac{2}{\sqrt{3}}\sqrt{U_2(z)}, \tag{14}$$

and hence $\{U_2 \le 3\theta^2/4\} \subseteq \{U \le \theta\} \subseteq \{U_2 \le \theta^2\}$.

Proof sketch. The lower bound is Proposition 6. For the upper bound, Lemma 3(i) gives $\min_k a_k \ge D/2$, and triangle inequality gives $a_k \le d_{\min} + D \le 3\min_k a_k$, so $\max/\min \le 3$. Among vectors in $[1,3]$, $\mathrm{RMS}/\mathrm{AM} \le 2/\sqrt{3}$ (achieved at a two-point distribution on $\{1,3\}$). Full derivation in Appendix B. □

Theorem 8 (Euclidean-ball geometry of U_2)**.** *Under A1′–A2 (balanced equal-norm code), for $\rho \in (0,1)$, define $t(\rho) := (1-\rho)/(1+\rho/(C-1)) \in (0,1)$ and $\gamma_\rho := 1/t(\rho)$. Then:*

(i) $U_2(z) = \dfrac{\|z\|^2 + R^2 - 2\alpha(z)}{\|z\|^2 + R^2 + \frac{2}{C-1}\alpha(z)}$.

(ii) $\{z : U_2(z) \le \rho\} = \bigcup_{j=1}^{C} \mathbb{B}_j(\rho)$ *where*

$$\mathbb{B}_j(\rho) := \left\{ z : \|z - \gamma_\rho s_j\|^2 \le R^2\left(\gamma_\rho^2 - 1\right)\right\}. \tag{15}$$

This set is compact for $\rho < 1$.

(iii) On $\|z\| = R$: $U_2(z) \le \rho \iff \exists j : \langle z, s_j\rangle \ge t(\rho)R^2$.

(iv) For $\theta \in [0,1)$,

$$\{U_2 \le \theta^2/2\} \subseteq \{U \le \theta\} \subseteq \{U_2 \le \theta^2\},$$

i.e., $\bigcup_j \mathbb{B}_j(\theta^2/2) \subseteq \{U \le \theta\} \subseteq \bigcup_j \mathbb{B}_j(\theta^2)$. Under A1″ (simplex), the inner set tightens to $\bigcup_j \mathbb{B}_j(3\theta^2/4)$ via Lemma 7.

Proof sketch. Part (i): expand $\|z - s_j\|^2 = \|z\|^2 + R^2 - 2\langle z, s_j\rangle$ and use $\sum_j s_j = 0$ (A1′). Part (ii): rearrange $U_2 \le \rho$ to $\alpha(z) \ge \frac{t(\rho)}{2}(\|z\|^2 + R^2)$; complete the square for each j. Part (iii): substitute $\|z\| = R$. Part (iv): the outer inclusion is Proposition 6. For the balanced-code inner inclusion, write $r = \|z\|$ and let $a_k = d_k(z)$ for $k \ne j^*(z)$. Then $a_k \le r + R$, so $\mathrm{RMS}^2 \le (r+R)\mathrm{AM}$, while part (i) gives $\mathrm{RMS}^2 = \nu(z) \ge r^2 + R^2$ because $\alpha(z) \ge 0$. Hence $\mathrm{RMS}/\mathrm{AM} \le (r+R)/\sqrt{r^2+R^2} \le \sqrt{2}$, i.e. $U \le \sqrt{2}\sqrt{U_2}$. The simplex sharpening uses Lemma 7. Full details in Appendix B. □

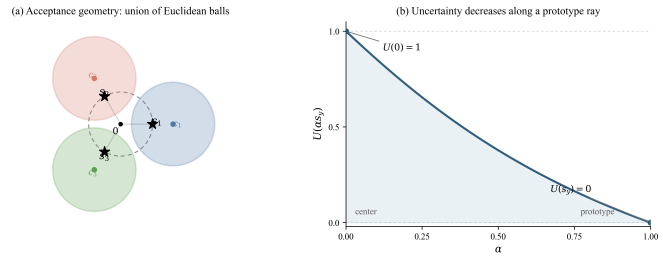

Figure 1: Illustration of the geometry for $C = 3$. Left: the U_2-sublevel set is a union of Euclidean balls $\mathbb{B}_j(\rho)$ centered along prototype directions, as characterized in Theorem 8(ii). Right: along a prototype ray $z = \alpha s_y$, the uncertainty decreases monotonically from the simplex center to the class prototype.

Key point: Theorem 8 requires only the balanced equal-norm condition A1′, *not* the full simplex A1″. The proof uses $\|s_j\| = R$ and $\sum_j s_j = 0$; equiangularity is never invoked. This means exact U_2-ball geometry is available in *every* dimension $d \ge 2$, including $d < C - 1$.

This theorem reveals that the U_2-sublevel set $\{U_2 \le \rho\}$ is a union of C (possibly overlapping) closed Euclidean balls, one per prototype. By the sandwich (Theorem 8(iv)), the operational U-acceptance region $\{U \le \theta\}$ lies between two compact prototype-aligned unions of balls. In the balanced equal-norm setting the global inner transfer uses $\theta^2/2$, while the simplex special case improves this to $3\theta^2/4$. The ball centers $\gamma_\rho s_j$ lie along the prototype directions at distance $\gamma_\rho R > R$ from the origin, and the ball radii grow as $\rho \to 1$. Combining with Theorem 2, this gives fully explicit compactness for U-acceptance regions for any balanced equal-norm code:

Corollary 9 (Compactness for U via U_2 transfer)**.** *Under A1′–A2, for any $\theta \in (0,1)$:*

$$\{U \le \theta\} \subseteq \bigcup_{j=1}^{C} \overline{B}\left(\gamma_{\theta^2} s_j,\ R\sqrt{\gamma_{\theta^2}^2 - 1}\right).$$

5.1 Sharp Dichotomy and the Simplex-Defect Parameter

The simplex-defect parameter $\lambda_j = \sqrt{B_j}/A_j$ introduced in A1 quantifies how far each class is from the one-distance simplex geometry. The following theorem shows that $d \ge C - 1$ is the phase transition for $\lambda_j = 1$, not for the rejection geometry itself.

Theorem 10 (Sharp dichotomy)**.** *Assume A1′ (balanced equal-norm). The following are equivalent:*

(i) $\lambda_j = 1$ *for every* j*;*

(ii) all pairwise distances $|s_i - s_j|$ *are equal;*

(iii) the prototypes form a regular simplex.

Consequently, if $C > d+1$*, then necessarily* $\max_j \lambda_j > 1$*.*

Proof. $\lambda_j = \sqrt{B_j}/A_j \geq 1$ with equality iff AM = RMS of the rival distances, i.e., all $|s_j - s_k|$ $(k \neq j)$ are equal. If this holds for every j, the off-diagonal distance matrix has each row constant, hence all entries are the same: the prototypes form a regular simplex. A regular simplex with C vertices requires affine dimension $C-1$, so $d \geq C-1$ is necessary. □

Theorem 11 (Local equivalence of U and $\sqrt{U_2}$)**.** *Assume A1 (distinct prototypes). If* $z \in B(s_j, r)$ *with* $r < \Delta_j/2$*, then* $j = j^*(z)$ *and*

$$\sqrt{U_2(z)} \leq U(z) \leq \frac{\sqrt{B_j}+r}{A_j - r}\sqrt{U_2(z)}. \tag{16}$$

Equivalently, if $r \leq \eta A_j$ *with* $0 < \eta < \Delta_j/(2A_j)$*:*

$$\sqrt{U_2(z)} \leq U(z) \leq \frac{\lambda_j + \eta}{1-\eta}\sqrt{U_2(z)}.$$

Proof. If $r < \Delta_j/2$, then j is uniquely nearest (as in Theorem 2). $\mu(z) \geq A_j - r$ gives $U(z) \leq r/(A_j - r)$. For U_2: $d_k(z) \leq |s_k - s_j| + r$ gives $\nu(z) \leq (\sqrt{B_j}+r)^2$ by Minkowski, so $\sqrt{U_2(z)} \geq r/(\sqrt{B_j}+r)$. Combining: $U(z) \leq \frac{\sqrt{B_j}+r}{A_j - r}\sqrt{U_2(z)}$. □

Interpretation. Near prototype s_j, the scores U and $\sqrt{U_2}$ are equivalent up to a factor that depends on λ_j. For $\lambda_j = 1$ (simplex case), the local comparison factor approaches 1 as $r \to 0$. For $\lambda_j > 1$ (low-dimensional regime), the factor degrades gracefully and quantifiably.

5.2 Universal Low-Dimensional Construction: The Regular C-gon

For every $d \geq 2$ and $C \geq 3$, the regular C-gon provides a concrete balanced equal-norm code:

$$s_j = R\big(\cos(2\pi j/C),\ \sin(2\pi j/C),\ 0, \ldots, 0\big), \quad j = 1, \ldots, C. \tag{17}$$

This satisfies $|s_j| = R$ and $\sum_j s_j = 0$. By symmetry, $A_j = A$, $B_j = B$, $\Delta_j = \Delta$, $\lambda_j = \lambda$ are all independent of j, with:

$$\Delta = 2R\sin\frac{\pi}{C}, \quad A = \frac{2R}{C-1}\cot\frac{\pi}{2C},$$
$$B = \frac{2CR^2}{C-1}, \quad \lambda = \frac{\sqrt{2C(C-1)}}{2\cot(\pi/(2C))}.$$

The acceptance region from Theorem 2 satisfies

$$\bigcup_{j=1}^{C} B\left(s_j,\ \min\left\{R\sin\frac{\pi}{C},\ \frac{2\theta R}{(1+\theta)(C-1)}\cot\frac{\pi}{2C}\right\}\right) \subseteq \{U \leq \theta\}$$

and

$$\{U \leq \theta\} \subseteq \bigcup_{j=1}^{C} \overline{B}\left(s_j,\ \frac{2\theta R}{(1-\theta)(C-1)}\cot\frac{\pi}{2C}\right).$$

At the same time, the exact U_2 ball geometry from Theorem 8 gives $\{U_2 \leq \rho\} = \bigcup_j \overline{B}(\gamma_\rho s_j, R\sqrt{\gamma_\rho^2 - 1})$.

A useful asymptotic for fixed $d = 2$ as $C \to \infty$: $A \sim 4R/\pi$, $\Delta \sim 2\pi R/C$. The guaranteed confident cores shrink because packing gets tighter, but the acceptance region remains compact with $O(R)$ outer scale—graceful degradation, not collapse.

6 False Acceptance Rate Bounds

Theorem 12 (KRR and conditional FAR bounds)**.** *Under A1′–A2 (balanced equal-norm code), and with additional assumptions as specified in each part:*
(i) KRR control. *Assume A5. Setting*

$$\theta_\varepsilon := \min\left\{1, \max_y \left(\mathbb{E}[U(Z) \mid Y = y] + \tau_y\sqrt{2\log(1/\varepsilon)}\right)\right\}$$

yields $\mathrm{KRR}(\theta_\varepsilon) \leq \varepsilon$*. In particular, the conservative common-scale choice* $\theta_\varepsilon = \min\left\{1, \max_y \left(\mathbb{E}[U(Z) \mid Y = y] + L_U\sigma\sqrt{2\log(1/\varepsilon)}\right)\right\}$ *also works.*
(ii) FAR via geometry (for $\theta \in [0,1)$***).*** $\mathrm{FAR}(\theta) \leq \mathbb{P}(\Phi(Z_U) \in \bigcup_j \mathbb{B}_j(\theta^2))$.
(iii) Under (U-Norm) + A6′ (for $\theta \in [0,1)$***):***

$$\mathrm{FAR}(\theta) \leq C\exp\left(-\frac{(d-1)t(\theta^2)^2}{2}\right) + C\,\delta_{\mathrm{cap}}. \tag{18}$$

If $\delta_{\mathrm{cap}} = 0$ *and* $\mathbb{P}(Z_U \neq 0) = 1$*, the exact-isotropy lower bound*

$$p_{\mathrm{cap}}(t(\theta^2/2); d) \leq \mathrm{FAR}(\theta)$$

also holds. Under the additional simplex assumption A1″, the sharper simplex-only lower bound

$$C\,p_{\mathrm{cap}}(t((\theta/3)^2); d) \leq \mathrm{FAR}(\theta)$$

holds.
(iv) Under (U-RadIso) + A6′ (for $\theta \in [0,1)$***):***

$$\mathrm{FAR}(\theta) \leq C\,\mathbb{E}\left[\exp\left(-\frac{d-1}{2}\min\{1, \tau_\theta(R_U)\}^2\right)\right], \tag{19}$$

where $\tau_\theta(r) := \frac{t(\theta^2)}{2} \cdot \frac{r^2+R^2}{rR}$ *for* $r > 0$*.*

Proof sketch. (i) follows from Lemma 5, total probability, and clipping at 1 when the raw threshold exceeds the score range. (ii) uses Proposition 6 and Theorem 8(ii). (iii): on the sphere, the upper bound comes from a union of spherical caps together with (U-Norm) and A6′. The exact-isotropy lower bound uses the inner inclusion from Theorem 8(iv); the stronger simplex-only lower bound additionally uses Lemma 7 and cap disjointness. (iv): condition on R_U, use isotropy of V_U, and apply A6′ pointwise. Full proofs in Appendix C. □

Interpretation. Equation (18) shows that, in the normalized-route analysis, the FAR upper bound for $U(\Phi(Z_U))$ is an exponential term plus an additive anisotropy floor $C\,\delta_{\text{cap}}$. Hence under exact isotropy ($\delta_{\text{cap}} = 0$), the normalized-route upper bound decays exponentially in d for any fixed $\theta < 1$; under the radial-isotropic model, equation (19) gives an exponential-in-d FAR upper bound for $U(Z_U)$. The term $t(\theta^2)$ quantifies how much of the sphere is *not* covered by the acceptance caps: smaller θ yields larger t, hence faster decay of the upper bound.

6.1 Why the FAR bound can be loose in practice

Although Theorem 12 provides exponential-in-d FAR scaling (the key qualitative insight), several factors contribute to conservatism in the bound, particularly in the prefactors and additive terms.

The proof uses a union bound over C spherical caps, yielding $\text{FAR}(\theta) \le C\exp(-(d-1)t(\theta^2)^2/2) + C\,\delta_{\text{cap}}$. The term δ_{cap} captures deviation of the true unknown-feature angular distribution from the ideal uniform sphere; when unknowns concentrate along a few learned directions, this term can be substantial. The union bound itself is conservative: when cap events overlap (which they often do in practice), summing C marginal probabilities overcounts their union, sometimes dramatically.

Beyond the union bound, the proof strategy itself introduces looseness. It assumes isotropy of unknown features in all directions $v \in S^{d-1}$, but learned embeddings can develop directions aligned with simplex geometry since features are trained with respect to those very prototypes. Additionally, the proof uses $\{U \le \theta\} \subseteq \{U_2 \le \theta^2\}$ (Proposition 6) to shift the analysis from the operational score U to the auxiliary score U_2, which enlarges the acceptance region and thus inflates the FAR bound. The spherical-cap approximation itself, namely $p_{\text{cap}}(t; d) \le \exp(-(d-1)t^2/2)$ from A6′, can be crude for moderate d and t close to 1.

Despite these sources of conservatism, the constants should be interpreted as measuring robustness rather than tight prediction. The main qualitative takeaway is the *exponential-in-d scaling*, which rigorously explains why even modest embedding dimensions can achieve strong FAR. The bound is most useful for reasoning about FAR trends across dimension, not for trusting absolute numerical predictions.

7 UCDSC Loss Analysis

We now analyze the three components of the UCDSC loss (6)–(8). Proposition 13 and Theorem 16 yield deterministic in-sample score bounds for the realized training known and auxiliary background features, while Proposition 15 identifies a local gradient mechanism for the L_u term.

Proposition 13 (Small L_{intra} implies a small average training U-score)**.** *Under A1 (distinct prototypes): if $L_{\text{intra}} \le \delta^2$, then $\frac{1}{n}\sum_i U(f_i) \le 2\delta/\Delta$, where $\Delta = \min_{i\ne j}|s_i - s_j|$. Under A1″ (simplex), this sharpens to $2\delta/D$.*

Proof. Since $d_{\min}(f_i) \le \|f_i - s_{y_i}\|$, the QM–AM inequality gives $\frac{1}{n}\sum_i d_{\min}(f_i) \le \delta$. By Theorem 2, $\mu(f_i) \ge \Delta/2$, so $U(f_i) \le 2d_{\min}(f_i)/\Delta$. Averaging gives the result. Under A1″, $\Delta = D$ and Lemma 3(i) gives $\mu(f_i) \ge D/2$. □

Proposition 14 (Zero L_o lower-bounds auxiliary background scores)**.** *Under A1 (distinct prototypes), assume every class $j \in [C]$ appears among $\{y_i\}_{i=1}^n$. If $L_o = 0$, then $U(f_k^{\text{bg}}) \ge \sqrt{m}/(D_{\max} + \sqrt{m})$ for all background samples f_k^{bg}, where $D_{\max} := \max_{i\ne j}|s_i - s_j|$. Under A1″ (simplex), $D_{\max} = D$ and we recover $U(f_k^{\text{bg}}) \ge \sqrt{m}/(D + \sqrt{m})$.*

Proof. $L_o = 0$ implies $d_{\min}(f_k^{\text{bg}}) \ge \sqrt{m}$. The triangle inequality gives $\mu(f_k^{\text{bg}}) \le d_{\min}(f_k^{\text{bg}}) + D_{\max}$, yielding the bound. □

Proposition 15 (Gradient structure of L_u)**.** *For $f_i \ne s_j$ for all j and assuming the nearest prototype $j^* = j^*(f_i)$ is unique (i.e. f_i does not lie on a bisector hyperplane):*

$$\nabla_{f_i} U(f_i) = \frac{1}{\mu}\left(\frac{f_i - s_{j^*}}{\|f_i - s_{j^*}\|} - \frac{U(f_i)}{C-1}\sum_{k\ne j^*}\frac{f_i - s_k}{\|f_i - s_k\|}\right). \tag{20}$$

Under gradient descent on the pointwise term $U(f_i)$ in L_u, the negative gradient $-\nabla_{f_i}U(f_i)$ has two interpretable components: the first moves f_i toward s_{j^*} (reducing $d_{\min}$), while the second moves f_i away from non-nearest prototypes (increasing μ). Both directions decrease $U(f_i)$ locally.

Theorem 16 (Conditional empirical separation for training known and auxiliary background samples)**.** *Under A1–A2, A4, assume every class $j \in [C]$ appears among $\{y_i\}_{i=1}^n$. Let $D_{\max} := \max_{i\ne j}|s_i - s_j|$. If at convergence $L_o = 0$, then:*

(i) Each training sample satisfies $U(f_i) \le 2\|f_i - s_{y_i}\|/\Delta$.

(ii) Each auxiliary background sample satisfies $U(f_k^{\mathrm{bg}}) \geq \sqrt{m}/(D_{\max} + \sqrt{m})$. In particular, if $\sqrt{m} \geq D_{\max}$, then $U(f_k^{\mathrm{bg}}) \geq 1/2$.

(iii) A separating threshold θ^* exists whenever

$$\max_i \|f_i - s_{y_i}\| < \frac{\Delta\sqrt{m}}{2(D_{\max} + \sqrt{m})}. \tag{21}$$

Under A1″ (simplex), $\Delta = D_{\max} = D$ and these reduce to the simplex-specific bounds.

Proof. (i): $d_{\min}(f_i) \leq \|f_i - s_{y_i}\|$ and $\mu(f_i) \geq \Delta/2$ (Theorem 2). (ii): Proposition 14. (iii): The known upper bound $2\max_i \|f_i - s_{y_i}\|/\Delta$ must be strictly less than the background lower bound; (21) is this condition. □

This is an in-sample statement about the realized training embeddings and the realized auxiliary background embeddings. It does not by itself imply rejection guarantees for future unknown classes, nor does it directly quantify open-space risk under a test-time unknown distribution. Within this scope, Theorem 16 shows how prototype closeness of known features and margin-based separation from the auxiliary background set induce score separation. Proposition 13 provides only an average training-score bound and does not by itself imply the uniform condition in (21). The role of L_u is different: Proposition 15 and Corollary 20 provide a local negative-gradient mechanism suggesting how minimizing L_u can move known features away from inter-class boundaries.

8 Practical Corollaries

Corollary 17 (Threshold selection)**.** *Under A1′–A2 and A5, setting*

$$\begin{aligned}\theta_\varepsilon &= \min\left\{1, \max_y \left(\mathbb{E}[U(Z) \mid Y{=}y] + \tau_y\sqrt{2\log(1/\varepsilon)}\right)\right\} \\ &= \min\left\{1, \max_y \left(\mathbb{E}[U(Z) \mid Y{=}y] + L_U\sigma_y\sqrt{2\log(1/\varepsilon)}\right)\right\}.\end{aligned} \tag{22}$$

guarantees population known-class acceptance $\geq 1-\varepsilon$. In practice, estimate $\mathbb{E}[U(Z) \mid Y{=}y]$ and the score-tail scales τ_y (or equivalently the rescaled parameters $\sigma_y = \tau_y/L_U$) from a validation set; for added finite-sample conservatism, use a safety margin or the DKW correction from Remark 22. In the simplex specialization A1″, one has $L_U = 4/D$.

Corollary 18 (Sufficient dimension for target FAR)**.** *Under (U-Norm) with $\delta_{\mathrm{cap}} = 0$, it suffices for $\mathrm{FAR}(\theta) \leq \eta$ that*

$$d \geq 1 + \frac{2\log(C/\eta)}{t(\theta^2)^2}. \tag{23}$$

For $C = 8$, $\theta = 0.5$, $\eta = 0.01$: $d \geq 1 + 2\log(800)/t(0.25)^2 \approx 27$. This is well within the embedding dimensions used in practice ($d \geq 64$).

Corollary 19 (Expand factor invariance under normalization)**.** *Under (U-Norm), the FAR bound (18) is independent of R. The expand factor R affects FAR only indirectly through the learned score-tail parameter σ. Without normalization (U-RadIso), the dependence on R enters through $\tau_\theta(R_U)$; whether increasing R tightens the bound depends on how the unknown-radius distribution of R_U is positioned relative to R.*

Corollary 20 (Local L_u descent mechanism)**.** *Gradient descent on L_u (i.e. the negative gradient $-\nabla_{f_i} U$, Proposition 15) simultaneously: (a) reinforces L_{intra} by pulling features toward the nearest prototype; and (b) pushes features away from inter-class boundaries, an effect that L_{intra} alone does not provide. This provides a local mechanism consistent with the empirical finding [7] that adding L_u alone ($\lambda_o = 0$, $\lambda_u \neq 0$) can improve OSCR [30] even without background data; it does not by itself prove global open-space reduction or test-time AUROC/OSCR gains.*

8.1 Practical Estimation of σ

Assumption 8 introduces the per-class upper-tail scales τ_y directly. The threshold rule (Theorem 12(i)) can be written in terms of τ_y or, equivalently, the rescaled score-tail parameters $\sigma_y = \tau_y/L_U$. We now give a concrete procedure to estimate these quantities from a labeled validation set.

Proposition 21 (Validation-based σ proxy)**.** *Let $\{(x_i, y_i)\}_{i=1}^N$ be a held-out validation set. Compute $z_i = f(x_i)$ and $u_i := U(z_i)$. For each class $y \in [C]$, let $I_y := \{i : y_i = y\}$, and define:*

*(a) **Empirical mean:** $\hat{m}_y := |I_y|^{-1} \sum_{i \in I_y} u_i$.*

*(b) **Empirical quantiles:** $\hat{q}_{1-\varepsilon,y} :=$ the $(1-\varepsilon)$-quantile of $\{u_i\}_{i \in I_y}$.*

*(c) **Upper-tail proxy:** For a grid $\mathcal{E} \subset (0,1)$ (e.g. $\mathcal{E} = \{0.1, 0.05, 0.01\}$),*

$$\hat{\tau}_y := \max_{\varepsilon \in \mathcal{E}} \frac{\left[\hat{q}_{1-\varepsilon,y} - \hat{m}_y\right]_+}{\sqrt{2\log(1/\varepsilon)}}, \tag{24}$$

with $[\cdot]_+$ as defined in Section 3.

Then back out $\hat{\sigma}_y = \hat{\tau}_y/L_U$, and set $\hat{\sigma} := \max_{y \in [C]} \hat{\sigma}_y$. In the simplex specialization A1″, this simplifies to $\hat{\sigma}_y = (D/4)\hat{\tau}_y$.

Justification. By Lemma 5, the population quantities satisfy

$$\frac{[q_{1-\varepsilon}(U(Z) \mid Y{=}y) - \mathbb{E}[U(Z) \mid Y{=}y]]_+}{\sqrt{2\log(1/\varepsilon)}} \leq \tau_y$$

for every $\varepsilon \in (0,1)$. Replacing the population mean and quantiles by their empirical counterparts over a grid $\mathcal{E}$ yields a practical proxy for τ_y. Even at the population

level this inversion is only a lower-envelope characterization, and with empirical inputs it can underestimate τ_y; consequently, the resulting threshold $\hat{\theta}_\varepsilon$ may be slightly optimistic (too small). In practice, adding a small safety margin or applying the DKW correction in Remark 22 compensates for this. The clipping $[\cdot]_+$ handles the (rare) case where the empirical quantile falls below the empirical mean due to finite-sample fluctuation.

Plugging in. With $\hat{\sigma}$ in hand, the threshold selection proxy (Corollary 17) becomes fully data-driven:

$$\begin{aligned}\hat{\theta}_\varepsilon &:= \min\left\{1, \max_{y\in[C]}\left(\hat{m}_y + \hat{\tau}_y\sqrt{2\log(1/\varepsilon)}\right)\right\} \\ &= \min\left\{1, \max_{y\in[C]}\left(\hat{m}_y + L_U\hat{\sigma}_y\sqrt{2\log(1/\varepsilon)}\right)\right\},\end{aligned} \tag{25}$$

which provides a practical, theorem-motivated alternative to grid-search threshold tuning.

Remark 22 (Finite-sample correction). For small validation sets ($|I_y| \lesssim 50$), the empirical quantiles $\hat{q}_{1-\varepsilon,y}$ are noisy. A Dvoretzky–Kiefer–Wolfowitz (DKW) correction can be applied: replace $\hat{q}_{1-\varepsilon,y}$ with $\hat{q}_{1-\varepsilon+\delta_n,y}$ where $\delta_n = \sqrt{\log(2/\alpha)/(2|I_y|)}$ for confidence level $1-\alpha$. This reduces quantile underestimation; a full high-probability guarantee $\hat{\sigma} \geq \sigma$ would also require controlling the mean estimation error $|\hat{m}_y - m_y|$ (e.g. via Hoeffding's inequality applied to the bounded random variable $U(Z) \in [0,1]$).

Remark 23 (Connection to expand factor). Under the generic A1/A1′ Lipschitz bound $L_U = 4C/((C-1)\Delta)$,

$$\hat{\sigma}_y = \frac{\hat{\tau}_y}{L_U} = \frac{(C-1)\Delta}{4C}\,\hat{\tau}_y.$$

Hence the dependence on the expand factor R enters through how the minimum separation Δ of the chosen prototype family scales with R. If the prototypes are obtained by scaling a fixed angular code, then Δ is linear in R. In the simplex specialization, $\Delta = D = R\sqrt{2C/(C-1)}$, recovering $\hat{\sigma}_y = (D/4)\hat{\tau}_y$. In practice, $\hat{\tau}_y$ (the directly observable tail scale of U-scores) is the more natural quantity. Under A5 alone, σ_y should be viewed as a rescaled score-tail parameter for U, not as a generic feature-space concentration parameter.

The complete estimation pipeline is summarized in Algorithm 1.

8.2 Practical Estimation of δ_{cap}

To apply the FAR bound (Theorem 12(iii)), one must estimate δ_{cap} from data. Collect a set of unknown embeddings $\{z_i\}_{i=1}^N$ and radially normalize each to radius R via $\Phi(z_i) = Rz_i/\|z_i\|$ (discard $z_i = 0$), so $\Phi(z_i)/R \in S^{d-1}$. Draw M random directions $v_m \sim \text{Unif}(S^{d-1})$, and choose thresholds t on a fine grid $\mathcal{T} \subset [0,1]$. For each t, compute the empirical cap tail $\hat{p}(t) = \frac{1}{NM}\sum_{i,m}\mathbf{1}\{\langle\Phi(z_i)/R, v_m\rangle \geq t\}$. Compute the corresponding spherical-cap probability $p_{\text{cap}}(t;d) = \mathbb{P}(\langle W, v\rangle \geq t)$ for $W \sim \text{Unif}(S^{d-1})$. Then define $\widehat{\delta}_{\text{cap}} = \max_{t\in\mathcal{T}}|\hat{p}(t) - p_{\text{cap}}(t;d)|$. Optionally, bootstrap over embeddings and directions to quantify finite-sample uncertainty.

Algorithm 1 Practical estimation of $\hat{\sigma}$ and $\hat{\theta}_\varepsilon$

Require: Trained model f, prototypes $\{s_1, \ldots, s_C\}$, validation set $\{(x_i, y_i)\}_{i=1}^N$, target known-recall miss rate ε, quantile grid $\mathcal{E}$

Ensure: Threshold proxy $\hat{\theta}_\varepsilon$, score-tail proxy $\hat{\sigma}$

```
Compute z_i ← f(x_i) and u_i ← U(z_i) for all i
Compute Δ ← min_{i≠j} ‖s_i − s_j‖,   L_U ← 4C/((C − 1)Δ)   ▷ if simplex: L_U = 4/D
for each class y ∈ [C] do
    I_y ← {i : y_i = y}
    m̂_y ← |I_y|^{−1} Σ_{i∈I_y} u_i
    for each ε′ ∈ E do
        Compute q̂_{1−ε′,y} from {u_i}_{i∈I_y}
    end for
    τ̂_y ← max_{ε′∈E} [q̂_{1−ε′,y} − m̂_y]_+ / √(2 log(1/ε′))
    σ̂_y ← τ̂_y / L_U
end for
σ̂ ← max_y σ̂_y
θ̂_ε ← min{1, max_y (m̂_y + τ̂_y √(2 log(1/ε)))}
return θ̂_ε, σ̂
```

Caveat. Assumption 9 (U-Norm) defines δ_{cap} as a worst-case (supremum-over-v) deviation, but the estimator averages over M random directions. For isotropic unknowns, all directions are equivalent and the estimator is consistent. For anisotropic unknowns, $\widehat{\delta}_{\text{cap}}$ is a Monte Carlo lower bound on the true δ_{cap}; a rigorous upper bound would require an ε-net or covering argument over S^{d-1} combined with a DKW-type bound over thresholds.

8.3 Numerical Illustrations

We report results from synthetic validation experiments implementing Algorithm 1 and the δ_{cap} estimator.

$\hat{\sigma}$ estimation. For a regular-simplex illustration with $C = 4$ classes, $R = 50$, $n = 30$ samples per class, and $\varepsilon = 0.01$, Algorithm 1 yields per-class results in Table 1. Here $D = 81.6$ and $L_U = 4/D$, so the reported $\hat{\sigma}_y$ values use the simplex specialization. The estimated threshold is $\hat{\theta}_{0.01} = 0.097$, well below 1, confirming that known features cluster tightly around prototypes.

$\widehat{\delta}_{\text{cap}}$ estimation. With $N = 1{,}500$ isotropic unknowns in $d = 16$, $R = 50$, and $M = 4{,}096$ random directions, we obtain $\widehat{\delta}_{\text{cap}} = 2.9 \times 10^{-4}$, confirming that the spherical-cap approximation is extremely accurate for isotropic features. For anisotropic unknown distributions, $\widehat{\delta}_{\text{cap}}$ would be larger.

FAR vs. dimension d (synthetic sanity check). To illustrate Theorem 12(iii), we sample 5,000 unknowns uniformly on S^{d-1} (scaled to $R = 50$) for $C = 4$, $\theta = 0.2$, and $d \in \{3, 5, 8, 16, 32, 64\}$. Table 2 and Figure 2 show

Table 1: Per-class $\hat{\sigma}$ estimation for the regular-simplex illustration ($C = 4$, $R = 50$, $D = 81.6$, $L_U = 4/D$). Values of $\hat{\sigma}_y$ are computed from unrounded $\hat{\tau}_y$; the displayed $\hat{\tau}_y$ are rounded to four decimal places.

Class	n	$\hat{m}_y$	$\hat{\tau}_y$	$\hat{\sigma}_y$
0	30	0.035	0.0103	0.210
1	30	0.037	0.0157	0.320
2	30	0.043	0.0177	0.362
3	30	0.038	0.0129	0.264
$\hat{\sigma} = \max_y \hat{\sigma}_y$			0.362	
$\hat{\theta}_{0.01}$			0.097	

Table 2: FAR vs. dimension d. Empirical FAR from 5,000 uniform-sphere unknowns and the Theorem 12(iii) bound ($C = 4$, $R = 50$, $\theta = 0.2$, $t(\theta^2) \approx 0.947$). Entries reported as $< 2 \times 10^{-4}$ correspond to zero observed false acceptances among 5,000 samples.

d	FAR (empirical)	FAR (bound)	$\log_{10}$(bound)
3	0.1068	1.000	0.0
5	0.0098	0.665	-0.18
8	0.0002	0.173	-0.76
16	$<2\times10^{-4}$	4.8×10^{-3}	-2.3
32	$<2\times10^{-4}$	3.6×10^{-6}	-5.4
64	$<2\times10^{-4}$	2.1×10^{-12}	-11.7

that empirical FAR drops rapidly; for $d \geq 16$, we observe zero false acceptances among 5,000 sampled unknowns, while the theoretical bound captures the expected exponential trend.

Prospective operational note: data-driven thresholding via $\hat{\sigma}$. Algorithm 1 provides a theorem-motivated threshold-selection rule for future implementations: given a target known-class miss rate ε, it selects $\hat{\theta}_\varepsilon$ from validation-set U-score quantiles via equation (25). This is intended as operational guidance rather than validation. Without a safety margin or DKW correction, the raw plug-in threshold can be slightly optimistic because $\hat{\tau}_y$ may underestimate the true tail scale. The rule controls known-class recall, not AUROC or OSCR directly. Datasets with larger intra-class spread would be expected to yield larger $\hat{\tau}_y$, if that spread is actually observed on validation data.

9 Empirical Support and Theory–Experiment Consistency

We now complement the theory with focused experiments on reproduced UCDSC baselines for BloodMNIST and DermaMNIST, and then relate the resulting measurements to the theoretical predictions. The aim of this section is not to claim exhaustive benchmark coverage, but to test the specific aspects of the framework that are most

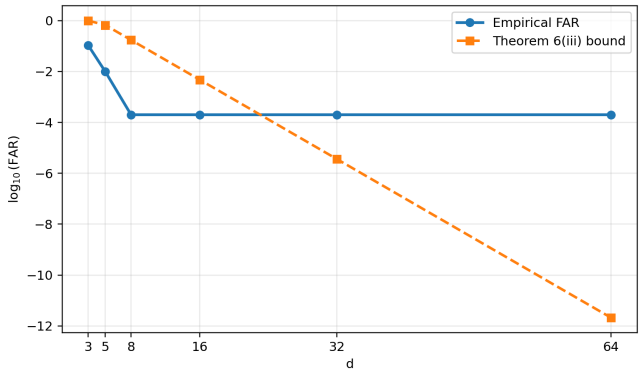

Figure 2: $\log_{10}$(FAR) vs. embedding dimension d for uniform-sphere unknowns ($C = 4$, $R = 50$, $\theta = 0.2$). Blue: empirical FAR, with zero-count cases displayed at the detection floor $1/5000 = 2 \times 10^{-4}$; red dashed: Theorem 12(iii) upper bound. The linear trend in log-scale is consistent with exponential-in-d FAR decay in this isotropic example.

Table 3: Reproduced UCDSC baseline results (mean $\pm$ std over splits).

Dataset	ACC	AUROC	OSCR	TNR	DTACC
BloodMNIST	98.242 ± 0.662	81.122 ± 8.230	80.544 ± 8.193	34.601 ± 18.037	76.820 ± 5.796
DermaMNIST	84.414 ± 6.389	67.373 ± 7.903	60.133 ± 11.355	14.545 ± 5.641	68.146 ± 3.255

directly operational: threshold choice, dimension sensitivity, and anisotropy diagnostics.

Reproduced baselines. Table 3 shows that the reproduced UCDSC baseline is much stronger on BloodMNIST than on DermaMNIST. This difference is visible in both ranking-style open-set metrics (AUROC and OSCR) and in operating-point metrics (TNR and DTACC), indicating that DermaMNIST provides a materially harder open-set regime. The basic pattern is therefore consistent with the paper's emphasis on operating conditions rather than closed-set accuracy alone.

Threshold calibration supports the operating-point analysis. The threshold results provide the clearest empirical support for the theory. At the recall-target operating point corresponding to approximately 95% known-class recall, the mean FAR is 0.652 ± 0.180 on BloodMNIST and 0.855 ± 0.056 on DermaMNIST. At the best-DTACC operating point, the mean FAR drops to 0.082 ± 0.026 and 0.244 ± 0.225, respectively, with the expected recall tradeoff. Thus the experiments confirm the central practical message of Theorem 12(i): threshold selection is a genuine operating decision, not a cosmetic post-processing choice. Controlling KRR forces a substantial FAR tradeoff, especially on the harder DermaMNIST setting.

Published UCDSC ablations remain consistent with the loss analysis. Corollary 20 suggests a local mechanism by which L_u can improve open-set behavior even without background data ($\lambda_o = 0$): its neg-

ative gradient both reinforces prototype attraction and pushes features away from inter-class boundaries. The UCDSC ablation on ASC (Table 4 in [7]) remains consistent with this mechanism: at $(\lambda_o, \lambda_u) = (0, 0.1)$, AUROC increases from 69.53% to 73.40%, and OSCR from 62.69% to 68.02%, close to the best scores at $(\lambda_o, \lambda_u) = (1, 2)$. Similarly, Theorem 16(iii) suggests that L_o should help more when known features are already well clustered, which remains qualitatively consistent with the stronger BloodMNIST behavior reported in the original UCDSC study.

A stronger low-dimension experiment supports dimension sensitivity, but not a full monotone FAR law. To test the dimension claim more directly, we reran BloodMNIST with a reduced embedding dimension of 32 over all five splits. The resulting baseline remains viable, but its open-set quality degrades relative to the default high-dimensional run: AUROC falls from 81.122 ± 8.230 to 78.108 ± 12.801, and OSCR from 80.544 ± 8.193 to 77.839 ± 12.824. The operating-point effect is stronger: at the best-DTACC threshold, the mean FAR increases from 0.082 ± 0.026 to 0.289 ± 0.184. This provides initial multi-split empirical support that embedding dimension materially affects rejection behavior. At the same time, the recall-target FAR comparison is noisier than the theorem's clean isotropic picture, so the current experiments support dimension sensitivity more strongly than they establish a sharp monotone empirical FAR law.

Anisotropy diagnostics remain empirically inconclusive. We also implemented the paper-style estimator of $\hat{\delta}_{\text{cap}}$ from Section 8.2. The resulting values are small and very similar across datasets: approximately 0.0167 ± 0.0077 on BloodMNIST and 0.0147 ± 0.0065 on DermaMNIST. Accepted-unknown alignment diagnostics likewise show only weak differences between all unknowns and accepted unknowns. Hence the current experiments do not isolate the anisotropy failure mode in a strong empirical way. This does not invalidate the warning in Section 6.1; rather, it indicates that the present diagnostics are not yet sufficient to explain the BloodMNIST–DermaMNIST gap through δ_{cap} alone.

Overall reading of the evidence. Taken together, the experiments support the theory most strongly at the threshold-selection and dimension-sensitivity levels. They show that the theory speaks directly to practical operating behavior, rather than merely to asymptotic geometry. By contrast, the anisotropy mechanism should still be presented as a theoretically motivated diagnostic warning whose empirical role remains open.

10 Conclusion

This paper develops, to our knowledge, the first comprehensive theoretical framework for simplex-ratio open-set recognition that works in *all* embedding dimensions, placing UCDSC and related methods on a rigorous geometric and probabilistic foundation. By introducing a hierarchy of prototype assumptions—from arbitrary distinct prototypes (A1) through balanced equal-norm codes (A1′) to the regular simplex (A1″)—we show that the core rejection geometry does not require the simplex assumption $d \geq C - 1$.

For arbitrary distinct prototypes, we establish that U is globally Lipschitz and that acceptance regions are compact finite unions of balls with explicit radii. For balanced equal-norm codes (available in every $d \geq 2$), we prove that U_2-sublevel sets are exact unions of Euclidean balls, and a sharp dichotomy identifies $d \geq C-1$ as the phase transition for simplex symmetry, not for the rejection geometry itself. Below this threshold, the simplex-defect parameter λ_j quantifies graceful degradation with explicit local comparison bounds. We prove FAR upper bounds that decay exponentially with embedding dimension, analyze the UCDSC loss terms, and derive practical data-driven threshold procedures.

We also complement the theory with focused reproduced experiments on BloodMNIST and DermaMNIST. These experiments show that threshold selection has a large and operationally meaningful effect on FAR and KRR, and that reducing the embedding dimension from the default high-dimensional setting to 32 measurably weakens open-set performance on BloodMNIST, especially at the best-DTACC operating point.

Limitations and future work. Key assumptions still deserve scrutiny. Assumption A5 (sub-Gaussian score tail) is an idealization that may not hold for poorly trained models or multi-modal class distributions; extending the analysis to weaker tail bounds is a natural next step. The RMS–AM ratio bound of $2/\sqrt{3}$ (Lemma 7) holds under A1″ and may be improvable for specific values of C; for general balanced equal-norm codes we use the coarser global factor $\sqrt{2}$ together with the local comparison of Theorem 11. The C-gon construction provides a concrete low-dimensional family, but optimizing the prototype configuration (e.g., via Grassmannian codes or group orbits) for specific (C, d) pairs is an open direction. On the empirical side, the anisotropy diagnostics remain inconclusive, and tightening the connection between theoretical FAR bounds and empirical rejection rates remains an open challenge that could guide future refinements of simplex-based OSR methods.

A Full Proof of Lemma 3

Recall $d_j(z) := \|z - s_j\|$, $d_{\min}(z) := \min_j d_j(z)$, and $\mu(z) := \frac{1}{C-1} \sum_{k \neq j^*(z)} d_k(z)$, where $j^*(z) \in \arg\min_j d_j(z)$.

(i) Pointwise bound $d_k(z) \geq D/2$ for $k \neq j^*(z)$, and hence $\mu(z) \geq D/2$. Fix z and let $j^* = j^*(z)$. Under A1″ (simplex), the common edge length is $D = \|s_{j^*} - s_k\|$ for all $k \neq j^*$. By the triangle inequality,

$$D = \|s_{j^*} - s_k\| \leq \|z - s_{j^*}\| + \|z - s_k\| = d_{\min}(z) + d_k(z).$$

Since $d_k(z) \geq d_{\min}(z)$, we have $D \leq 2d_k(z)$, hence $d_k(z) \geq D/2$ for all $k \neq j^*$. Averaging over $k \neq j^*$ yields $\mu(z) \geq D/2$.

(ii) Global Lipschitzness of $U(z) = d_{\min}(z)/\mu(z)$. First, $d_{\min}$ is 1-Lipschitz because it is the pointwise minimum of 1-Lipschitz maps:

$$|d_{\min}(z) - d_{\min}(z')| \leq \|z - z'\|.$$

Next, define for each $j \in [C]$ the function

$$\mu_j(z) := \frac{1}{C-1} \sum_{k \neq j} d_k(z).$$

Each μ_j is 1-Lipschitz as an average of $C-1$ functions each 1-Lipschitz, with coefficients $1/(C-1)$ summing to 1. Moreover, because removing the smallest distance maximizes the remaining average,

$$\mu(z) = \mu_{j^*(z)}(z) = \max_{j \in [C]} \mu_j(z).$$

The pointwise maximum of 1-Lipschitz functions is 1-Lipschitz, so $|\mu(z) - \mu(z')| \leq \|z - z'\|$.

Finally, note $d_{\min}(z) \leq \mu(z)$ and $d_{\min}(z') \leq \mu(z')$ (each non-nearest distance is $\geq d_{\min}$). Using the identity (for $0 \leq a \leq b$, $0 \leq a' \leq b'$):

$$\begin{aligned} \left| \frac{a}{b} - \frac{a'}{b'} \right| &= \frac{|ab' - a'b|}{bb'} \\ &\leq \frac{|a - a'|b' + a'|b - b'|}{bb'} \\ &\leq \frac{|a - a'| + |b - b'|}{\min\{b, b'\}}, \end{aligned}$$

where the last step uses $a' \leq b'$. Applying (i) so that $\min\{\mu(z), \mu(z')\} \geq D/2$, we obtain

$$\begin{aligned} |U(z) - U(z')| &\leq \frac{|d_{\min}(z) - d_{\min}(z')| + |\mu(z) - \mu(z')|}{D/2} \\ &\leq \frac{2}{D}\Big(\|z - z'\| + \|z - z'\|\Big) \\ &= \frac{4}{D}\|z - z'\|. \end{aligned}$$

Thus U is globally Lipschitz with constant $L_U = 4/D$.

B Full Proof of Theorem 8

Full derivation of Lemma 7 (U–U_2 equivalence). Fix $z \in \mathbb{R}^d$ and write $j^* = j^*(z)$. Let $a_k := d_k(z)$ for $k \neq j^*$ denote the $C-1$ non-nearest distances. Define

$$\mathrm{AM} := \frac{1}{C-1} \sum_{k \neq j^*} a_k, \qquad \mathrm{RMS} := \sqrt{\frac{1}{C-1} \sum_{k \neq j^*} a_k^2}.$$

By construction, $U(z) = d_{\min}(z)/\mathrm{AM}$ and $\sqrt{U_2(z)} = d_{\min}(z)/\mathrm{RMS}$, hence $U(z)/\sqrt{U_2(z)} = \mathrm{RMS}/\mathrm{AM}$.

Lower bound. Always AM $\leq$ RMS (by Cauchy–Schwarz), so $\sqrt{U_2(z)} \leq U(z)$.

Upper bound via bounded spread. Let $m := \min_{k \neq j^*} a_k$ and $M := \max_{k \neq j^*} a_k$. We claim $M/m \leq 3$. Indeed, for any $k \neq j^*$:

$$a_k = d_k(z) \leq d_{j^*}(z) + \|s_k - s_{j^*}\| = d_{\min}(z) + D \leq m + D,$$

since $m \geq d_{\min}(z)$. Also, by Lemma 3(i), $m \geq D/2$, hence $D \leq 2m$ and $a_k \leq m + 2m = 3m$. Taking the maximum over k gives $M \leq 3m$.

Now scale by m so that $x_k := a_k/m \in [1, 3]$. Then

$$\frac{\mathrm{RMS}}{\mathrm{AM}} = \frac{\sqrt{\frac{1}{C-1} \sum x_k^2}}{\frac{1}{C-1} \sum x_k}.$$

For a fixed mean, the second moment over $[1, 3]$ is maximized when all mass lies at the interval endpoints. Hence the worst case for $\sqrt{\mathbb{E}[x^2]}/\mathbb{E}[x]$ is attained by a two-point distribution on $\{1, 3\}$. Writing p for the fraction of entries equal to 3, this yields

$$\begin{aligned} \frac{\mathrm{RMS}}{\mathrm{AM}} &\leq \max_{p \in [0,1]} \frac{\sqrt{(1-p) \cdot 1^2 + p \cdot 3^2}}{(1-p) \cdot 1 + p \cdot 3} \\ &= \max_{p \in [0,1]} \frac{\sqrt{1 + 8p}}{1 + 2p} = \frac{2}{\sqrt{3}}, \end{aligned}$$

achieved at $p = 1/4$. Therefore $U(z) \leq \frac{2}{\sqrt{3}} \sqrt{U_2(z)}$. The set inclusions now follow in two steps:

$$U_2 \leq \frac{3\theta^2}{4} \;\Rightarrow\; U \leq \frac{2}{\sqrt{3}} \sqrt{U_2} \leq \frac{2}{\sqrt{3}} \cdot \frac{\theta\sqrt{3}}{2} = \theta,$$

and

$$U \leq \theta \;\Rightarrow\; \sqrt{U_2} \leq U \leq \theta \;\Rightarrow\; U_2 \leq \theta^2.$$

□

Part (i). $\|z - s_j\|^2 = \|z\|^2 + R^2 - 2\langle z, s_j \rangle$. Hence $d_{\min}^{(2)} = \|z\|^2 + R^2 - 2\alpha(z)$. Since $\sum_j s_j = 0$: $\sum_j \|z - s_j\|^2 = C\|z\|^2 + CR^2$. Therefore:

$$\begin{aligned} \nu(z) &= \frac{1}{C-1}(C\|z\|^2 + CR^2 - \|z\|^2 - R^2 + 2\alpha(z)) \\ &= \|z\|^2 + R^2 + \frac{2}{C-1}\alpha(z). \end{aligned}$$

Note $\alpha(z) \geq 0$ (since the C values $\langle z, s_j \rangle$ sum to zero), so $\nu(z) > 0$.

Part (ii). $U_2(z) \leq \rho$ rearranges to $\alpha(z) \geq \frac{t(\rho)}{2}(\|z\|^2 + R^2)$, i.e. $\exists j : \langle z, s_j \rangle \geq \frac{t}{2}(\|z\|^2 + R^2)$. Rearranging: $\|z\|^2 - \frac{2}{t}\langle z, s_j \rangle + R^2 \leq 0$. Completing the square: $\|z - s_j/t\|^2 = \|z\|^2 - \frac{2}{t}\langle z, s_j \rangle + R^2/t^2$, so the condition becomes $\|z - s_j/t\|^2 \leq R^2(1/t^2 - 1) =: r_j^2$. This is exactly $z \in \mathbb{B}_j(\rho)$. Since $t \in (0, 1)$, each ball has finite radius; the union is compact.

Part (iii). On $\|z\| = R$: $\frac{t}{2}(R^2 + R^2) = tR^2$.

Balanced-code comparison used in Part (iv). Assume A1′ and keep the notation

$$\mathrm{AM} := \frac{1}{C-1}\sum_{k\neq j^*} a_k, \qquad \mathrm{RMS} := \sqrt{\frac{1}{C-1}\sum_{k\neq j^*} a_k^2}.$$

Let $r := \|z\|$. Since $\|s_k\| = R$ for every k, triangle inequality gives $a_k \le r + R$, hence

$$\mathrm{RMS}^2 = \frac{1}{C-1}\sum_{k\neq j^*} a_k^2 \le (r+R)\frac{1}{C-1}\sum_{k\neq j^*} a_k = (r+R)\mathrm{AM}.$$

By Part (i),

$$\mathrm{RMS}^2 = \nu(z) = r^2 + R^2 + \frac{2}{C-1}\alpha(z) \ge r^2 + R^2,$$

because $\alpha(z) \ge 0$. Therefore

$$\frac{U(z)}{\sqrt{U_2(z)}} = \frac{\mathrm{RMS}}{\mathrm{AM}} \le \frac{r+R}{\mathrm{RMS}} \le \frac{r+R}{\sqrt{r^2+R^2}} \le \sqrt{2}.$$

Thus $U(z) \le \sqrt{2}\sqrt{U_2(z)}$ under A1′.

Part (iv). By Proposition 6, $\{U \le \theta\} \subseteq \{U_2 \le \theta^2\}$. For the inner inclusion under A1′,

$$U_2 \le \theta^2/2 \;\Rightarrow\; U \le \sqrt{2}\sqrt{U_2} \le \sqrt{2}\cdot\frac{\theta}{\sqrt{2}} = \theta.$$

Hence $\{U_2 \le \theta^2/2\} \subseteq \{U \le \theta\}$. Under the additional simplex assumption A1″, Lemma 7 improves this to $\{U_2 \le 3\theta^2/4\} \subseteq \{U \le \theta\}$. Apply part (ii) to the relevant sublevel sets.

C Full Proof of Theorem 12

Part (i): KRR control. Let $m_y := \mathbb{E}[U(Z) \mid Y = y]$ and $t_{\varepsilon,y} := \tau_y\sqrt{2\log(1/\varepsilon)}$. By Lemma 5, any threshold $\bar\theta_\varepsilon \ge \max_y(m_y + t_{\varepsilon,y})$ satisfies $\mathbb{P}(U(Z) > \bar\theta_\varepsilon \mid Y = y) \le \varepsilon$ for each y. If $\bar\theta_\varepsilon \le 1$, take $\theta_\varepsilon := \bar\theta_\varepsilon$ and average:

$$\mathrm{KRR}(\theta_\varepsilon) \le \sum_y \mathbb{P}(Y = y)\cdot\varepsilon = \varepsilon.$$

If $\bar\theta_\varepsilon > 1$, take $\theta_\varepsilon := 1$; since $U(Z) \in [0,1]$, this gives $\mathrm{KRR}(\theta_\varepsilon) = 0 \le \varepsilon$. The common-scale version follows from the conservative bound $t_{\varepsilon,y} \le \tau\sqrt{2\log(1/\varepsilon)} = L_U\sigma\sqrt{2\log(1/\varepsilon)}$.

Part (ii): FAR via geometry. $\sqrt{U_2} \le U$ (Proposition 6) $\Rightarrow \{U \le \theta\} \subseteq \{U_2 \le \theta^2\}$. Apply Theorem 8(ii).

Part (iii): FAR under (U-Norm) + A6′ for $\theta \in [0,1)$. *Step 1: Reduce to spherical caps.* By the convention $\Phi(0) := 0$ in Assumption 9, and since $\theta < 1$, the point 0 is rejected because $U(0) = 1$. Thus the acceptance event $\{U \le \theta\}$ forces $Z_U \neq 0$, where $\|\Phi(Z_U)\| = R$. By Theorem 8(iii), on the sphere $\|z\| = R$:

$$U_2(z) \le \theta^2 \iff \exists\, j \in [C] : \langle z, s_j\rangle \ge t(\theta^2)R^2.$$

Let $\widetilde{Z} = \Phi(Z_U)/R \in S^{d-1}$ and $v_j = s_j/R$. The condition becomes $\exists\, j : \langle\widetilde{Z}, v_j\rangle \ge t(\theta^2)$.

Step 2: Union bound. Since acceptance forces $Z_U \neq 0$ (Step 1), $\mathrm{FAR}(\theta) \le \mathbb{P}(U_2(\Phi(Z_U)) \le \theta^2)$ by Proposition 6. On $\|z\| = R$, this is $\mathbb{P}(\bigcup_{j=1}^C\{\langle\widetilde{Z}, v_j\rangle \ge t\} \mid Z_U \neq 0)\mathbb{P}(Z_U \neq 0)$. Since $\mathbb{P}(Z_U \neq 0) \le 1$:

$$\begin{aligned}\mathrm{FAR}(\theta) &\le \mathbb{P}\left(\bigcup_{j=1}^C\{\langle\widetilde{Z}, v_j\rangle \ge t\} \;\middle|\; Z_U \neq 0\right)\\ &\le \sum_{j=1}^C \mathbb{P}(\langle\widetilde{Z}, v_j\rangle \ge t \mid Z_U \neq 0). \quad \text{(union bound)}\end{aligned}$$

Step 3: Apply (U-Norm) and A6′. By (U-Norm), for each j:

$$\mathbb{P}(\langle\widetilde{Z}, v_j\rangle \ge t \mid Z_U \neq 0) \le p_{\mathrm{cap}}(t; d) + \delta_{\mathrm{cap}}.$$

This is precisely where δ_{cap} enters: it measures how far the true angular distribution of normalized unknowns deviates from the ideal uniform sphere. By A6′, $p_{\mathrm{cap}}(t; d) \le \exp(-(d-1)t^2/2)$. Summing over j:

$$\mathrm{FAR}(\theta) \le C\big(\exp(-(d-1)t(\theta^2)^2/2) + \delta_{\mathrm{cap}}\big).$$

Step 4: Lower bounds under exact isotropy. Assume $\delta_{\mathrm{cap}} = 0$ and $\mathbb{P}(Z_U \neq 0) = 1$. By Theorem 8(iv), under A1′ we have $\{U_2 \le \theta^2/2\} \subseteq \{U \le \theta\}$. Hence

$$\mathbb{P}(U_2(\Phi(Z_U)) \le \theta^2/2) \le \mathrm{FAR}(\theta).$$

On the sphere $\|z\| = R$, the event $\{U_2(z) \le \theta^2/2\}$ is the union of caps

$$\bigcup_{j=1}^C\{\langle\widetilde{Z}, v_j\rangle \ge t(\theta^2/2)\}.$$

The probability of this union is at least the probability of any one cap, so exact isotropy gives

$$p_{\mathrm{cap}}(t(\theta^2/2); d) \le \mathrm{FAR}(\theta).$$

If in addition A1″ holds, then Lemma 7 implies the smaller inner inclusion $\{U_2 \le (\theta/3)^2\} \subseteq \{U \le \theta\}$. The corresponding cap threshold satisfies $t((\theta/3)^2) \ge t(1/9) = 8(C-1)/(9C-8) \ge 4/5 > 1/\sqrt{2}$. Thus each cap half-angle is strictly less than $\pi/4$. Simplex cap centers satisfy $\langle v_i, v_j\rangle = -1/(C-1) \le 0$, so their pairwise angular separation is at least $\pi/2$. Hence these simplex caps are pairwise disjoint, and exact isotropy yields the sharper simplex-only bound

$$C\,p_{\mathrm{cap}}(t((\theta/3)^2); d) \le \mathrm{FAR}(\theta).$$

Part (iv): FAR under (U-RadIso) + A6′ for $\theta \in [0,1)$. Under (U-RadIso), $\Phi = \mathrm{id}$ (Assumption 9), so $\mathrm{FAR}(\theta) = \mathbb{P}(U(Z_U) \le \theta)$. By Proposition 6, $\{U \le \theta\} \subseteq \{U_2 \le \theta^2\}$, so $\mathrm{FAR}(\theta) \le \mathbb{P}(U_2(Z_U) \le \theta^2)$. Write

$Z_U = R_U V_U$ with $R_U = \|Z_U\|$ and $V_U \mid R_U$ uniform on S^{d-1}. For $R_U = r > 0$: $\langle Z_U, s_j \rangle \geq \frac{t(\theta^2)}{2}(r^2 + R^2) \iff \langle V_U, v_j \rangle \geq \tau_\theta(r)$ where $\tau_\theta(r) = \frac{t(\theta^2)}{2} \cdot \frac{r^2+R^2}{rR}$. For $R_U = 0$: $Z_U = 0$, and since $\theta < 1$ we have $U(0) = 1 > \theta$, so the point is rejected. Condition on $R_U = r$, apply the union bound over j, use isotropy of $V_U \mid R_U$, and apply A6′:

$$\mathrm{FAR}(\theta) \leq C\,\mathbb{E}\left[\exp\left(-\frac{d-1}{2}\min\{1, \tau_\theta(R_U)\}^2\right)\right].$$

The clipping $\min\{1, \cdot\}$ ensures p_{cap} receives an argument in $[0, 1]$; it is necessary when R_U is far from R, making $\tau_\theta(R_U) > 1$.

D Proof of Theorem 16

Recall that Theorem 16 is stated under A1–A2 and uses

$$\Delta := \min_{i \neq j} \|s_i - s_j\|, \qquad D_{\max} := \max_{i \neq j} \|s_i - s_j\|.$$

Part (i). $d_{\min}(f_i) \leq \|f_i - s_{y_i}\|$ (the nearest prototype is at most as far as the true one). By the proof of Theorem 2, $\mu(f_i) \geq \Delta/2$. Hence

$$U(f_i) \leq \frac{2\|f_i - s_{y_i}\|}{\Delta}.$$

Part (ii). $L_o = 0$ implies $\|f_k^{\mathrm{bg}} - s_j\| \geq \sqrt{m}$ for all j, k (Proposition 14). Hence $d_{\min}(f_k^{\mathrm{bg}}) \geq \sqrt{m}$. The triangle inequality gives $\mu(f_k^{\mathrm{bg}}) \leq d_{\min}(f_k^{\mathrm{bg}}) + D_{\max}$, and therefore

$$U(f_k^{\mathrm{bg}}) \geq \frac{\sqrt{m}}{D_{\max} + \sqrt{m}}.$$

If $\sqrt{m} \geq D_{\max}$, then $\sqrt{m}/(D_{\max} + \sqrt{m}) \geq D_{\max}/(2D_{\max}) = 1/2$.

Part (iii). Part (i) gives the known-feature upper bound $2\max_i \|f_i - s_{y_i}\|/\Delta$. Part (ii) gives the background lower bound $\sqrt{m}/(D_{\max} + \sqrt{m})$. Any threshold strictly between these two quantities separates the two sets, and the condition that such a threshold exists is exactly (21). In the simplex specialization A1″, one has $\Delta = D_{\max} = D$, recovering the simpler formulas stated in the main text.

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